

PROBABILITY DISTRIBUTION FUNCTIONS

Session 40 — PDF, PMF & CDF · Random Variables · Density Estimation

Random Variables

Discrete vs Continuous

PMF

PDF

CDF

Density Estimation

KDE

Session Notes — Complete Reference

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1 • What is a Random Variable?

A **random variable** is a function that maps outcomes from a random experiment's sample space to numerical values. It's the bridge between real-world events and mathematical analysis.

1.1 Formal Definition

DEFINITION

$X : S \rightarrow \mathbb{R}$ A random variable X is a function from the sample space S to the set of real numbers $\mathbb{R}$.

1.2 Why Do We Need Random Variables?

- **Quantification:** Converts qualitative outcomes (Heads/Tails) into numbers (1/0)
- **Mathematical operations:** Enables computing means, variances, and probabilities
- **Modelling:** Provides the foundation for all probability distributions used in ML
- **Universality:** Different experiments can share the same mathematical structure

Example — Coin Toss Experiment

Experiment: Toss a coin 3 times.

Sample space S : {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}

Random variable X = "Number of heads"

X maps: HHH $\rightarrow$ 3, HHT $\rightarrow$ 2, HTH $\rightarrow$ 2, THH $\rightarrow$ 2, HTT $\rightarrow$ 1, THT $\rightarrow$ 1, TTH $\rightarrow$ 1, TTT $\rightarrow$ 0

So X can take values {0, 1, 2, 3} — it converts outcomes to numbers.

Notation Convention

- **Capital X :** Represents the random variable itself (the function)
- **Lowercase x :** Represents a specific value the random variable can take
- **$P(X = x)$:** "The probability that the random variable X equals the value x "

2 • Types of Random Variables

Discrete Random Variable

Takes a **countable** number of distinct values (finite or countably infinite).

Think: You can *count* the outcomes

Examples:

- Number of heads in 10 coin tosses (0–10)
- Dice roll result (1–6)
- Number of customers arriving per hour
- Number of defective items in a batch

Described by: PMF (Probability Mass Function)

Continuous Random Variable

Takes **uncountably infinite** values within a range or interval.

Think: You *measure* the outcomes

Examples:

- Height of a person (170.234... cm)
- Temperature (36.7823... °C)
- Time to complete a task (4.56789... min)
- Weight of a product (500.123... g)

Described by: PDF (Probability Density Function)

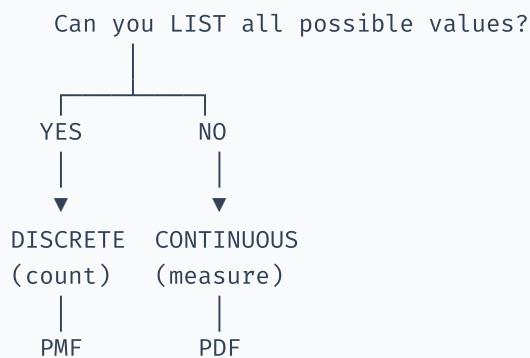
Critical Difference

For **discrete** variables: $P(X = \text{specific value})$ can be $> 0 \rightarrow$ we can assign probability to each value.

For **continuous** variables: $P(X = \text{any exact value}) = 0 \rightarrow$ probability exists only over *intervals*.

2.1 How to Identify: Quick Decision Test

Decision Flowchart



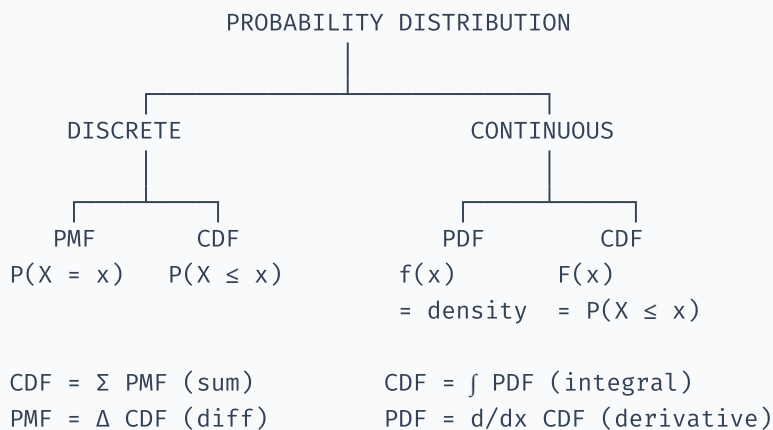
3 · Probability Distributions — The Big Picture

A **probability distribution** describes how the probabilities are spread (distributed) across all possible values of a random variable. It is the complete mathematical description of randomness.

3.1 The Three Core Functions

FUNCTION	FULL NAME	APPLIES TO	WHAT IT TELLS YOU
PMF	Probability Mass Function	Discrete RVs	$P(X = x)$ — Probability at each specific value
PDF	Probability Density Function	Continuous RVs	$f(x)$ — Density at each point (NOT probability directly)
CDF	Cumulative Distribution Function	Both	$F(x) = P(X \leq x)$ — Cumulative probability up to x

The Complete Map



3.2 Why Distributions Matter in ML

- **Feature modelling:** Understanding if features follow known distributions (Normal, Poisson, etc.)
- **Assumption checking:** Many ML algorithms assume normally distributed data
- **Anomaly detection:** Points in the extreme tails of a distribution may be anomalies
- **Generative models:** Learn the underlying distribution to generate new data (GANs, VAEs)
- **Bayesian inference:** Prior and posterior distributions are central to Bayesian ML

4 • Probability Mass Function (PMF)

The **PMF** is the probability function for **discrete random variables**. It assigns a probability to *each individual value* that the random variable can take.

4.1 Formal Definition

PMF DEFINITION

$P(X = x) = p(x)$ for each possible value x where $p(x) \geq 0$ for all x , and $\sum p(x) = 1$ (over all possible values)

4.2 Properties of PMF

1. **Non-negativity:** $p(x) \geq 0$ for every possible value x
2. **Sum to 1:** The sum of all probabilities equals exactly 1: $\sum p(x_i) = 1$
3. **Individual probability:** Each $p(x)$ gives the exact probability of that specific outcome
4. **Zero elsewhere:** $p(x) = 0$ for values that are impossible

4.3 Examples of PMF

Example 1 — Fair Die Roll

Random variable **X** = face value when rolling a fair die

X	1	2	3	4	5	6
$P(X = x)$	1/6	1/6	1/6	1/6	1/6	1/6

✓ Each $p(x) \geq 0$ ✓ Sum = $6 \times (1/6) = 1$

Example 2 — Two Coin Tosses (X = number of Heads)

Sample space: {HH, HT, TH, TT} **X values:** {0, 1, 2}

X	0 (TT)	1 (HT, TH)	2 (HH)
$P(X = x)$	$1/4 = 0.25$	$2/4 = 0.50$	$1/4 = 0.25$

✓ $0.25 + 0.50 + 0.25 = 1.00$

4.4 Visualising PMF

A PMF is visualised as a **bar chart / stem plot** — discrete spikes at each possible value. The height of each bar/spike represents the probability. Unlike a histogram, there is **no width** to the bars (they represent individual points, not ranges).

5 • CDF for Discrete Random Variables

The **Cumulative Distribution Function (CDF)** gives the probability that the random variable is **less than or equal to** a given value.

5.1 Definition

CDF FOR DISCRETE RANDOM VARIABLES

$$F(x) = P(X \leq x) = \sum p(x_i) \text{ for all } x_i \leq x$$

5.2 Example — CDF from the Coin Toss PMF

X	PMF: $P(X = x)$	CDF: $F(x) = P(X \leq x)$
0	0.25	0.25
1	0.50	$0.25 + 0.50 = 0.75$
2	0.25	$0.75 + 0.25 = 1.00$

5.3 Properties of Discrete CDF

- **Non-decreasing:** $F(x)$ can only stay the same or increase as x increases
- **Step function:** Jumps occur at each value where $P(X = x) > 0$
- **Range:** $0 \leq F(x) \leq 1$
- **Limits:** $F(-\infty) = 0$ and $F(+\infty) = 1$
- **Recovery:** $P(X = x) = F(x) - F(x^-)$ (jump at x)

5.4 Using CDF to Find Probabilities

USEFUL CDF IDENTITIES

$$P(X \leq b) = F(b) \quad P(X > a) = 1 - F(a) \quad P(a < X \leq b) = F(b) - F(a)$$

6 • Probability Density Function (PDF)

The **PDF** is the probability function for **continuous random variables**. Unlike the PMF, the PDF does *not* give probability directly — it gives **probability density**.

6.1 The Key Insight

⚠ Crucial Understanding

For a continuous random variable, the probability of taking any **exact** value is **ZERO**.

$P(X = 170.0000...) = 0$ (there are infinitely many possible values in any interval)

Instead, we can only compute probability over an **interval**: $P(a \leq X \leq b) = \int_a^b f(x) dx = \text{area under the curve}$.

6.2 Formal Definition

PDF DEFINITION

$f(x)$ = probability density at point x $P(a \leq X \leq b) = \int_a^b f(x) dx$ = Area under $f(x)$ from a to b

6.3 Properties of PDF

1. **Non-negativity**: $f(x) \geq 0$ for all x
2. **Total area = 1**: $\int_{-\infty}^{+\infty} f(x) dx = 1$
3. **$f(x)$ is NOT probability**: $f(x)$ can be greater than 1! (e.g., $\text{Uniform}(0, 0.5)$ has $f(x) = 2$)
4. **Point probability = 0**: $P(X = c) = \int^c f(x) dx = 0$
5. **Consequence**: $P(a \leq X \leq b) = P(a < X < b)$ (endpoints don't matter for continuous RVs)

Example — Uniform Distribution on [0, 1]

PDF: $f(x) = 1$ for $0 \leq x \leq 1$, and $f(x) = 0$ elsewhere

Find $P(0.3 \leq X \leq 0.7)$:

$$= \int_{0.3}^{0.7} 1 \cdot dx = [x]_{0.3}^{0.7} = 0.7 - 0.3 = \mathbf{0.4}$$

Verify total area: $\int_0^1 1 \cdot dx = 1 - 0 = 1 \checkmark$

 Example — Triangular PDF

PDF: $f(x) = 2x$ for $0 \leq x \leq 1$, and $f(x) = 0$ elsewhere

Check it's valid: $\int_0^1 2x \cdot dx = [x^2]_0^1 = 1 - 0 = 1 \checkmark$

Find $P(X \leq 0.5)$:

$$= \int_0^{0.5} 2x \cdot dx = [x^2]_0^{0.5} = 0.25$$

Only 25% of the probability mass lies in the first half — the distribution is skewed towards 1.

7 · CDF for Continuous Random Variables

7.1 Definition

CDF FOR CONTINUOUS RANDOM VARIABLES

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

7.2 Properties of Continuous CDF

- **Continuous function:** No jumps (unlike discrete CDF), smooth and continuous everywhere
- **Non-decreasing:** $F(a) \leq F(b)$ when $a \leq b$
- **Range:** $0 \leq F(x) \leq 1$
- **Limits:** $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow +\infty} F(x) = 1$
- **S-shaped:** Typically resembles an S-curve (sigmoid) for bell-shaped PDFs

Example — CDF for $f(x) = 2x$ on $[0, 1]$

CDF: $F(x) = \int_0^x 2t \cdot dt = [t^2]_0^x = x^2$ for $0 \leq x \leq 1$

Complete CDF:

$$F(x) = 0 \quad \text{for } x < 0$$

$$F(x) = x^2 \quad \text{for } 0 \leq x \leq 1$$

$$F(x) = 1 \quad \text{for } x > 1$$

Verify: $F(0) = 0$ ✓ $F(1) = 1$ ✓ $F(0.5) = 0.25$ (matches earlier calculation ✓)

8 • Relationship Between PDF & CDF

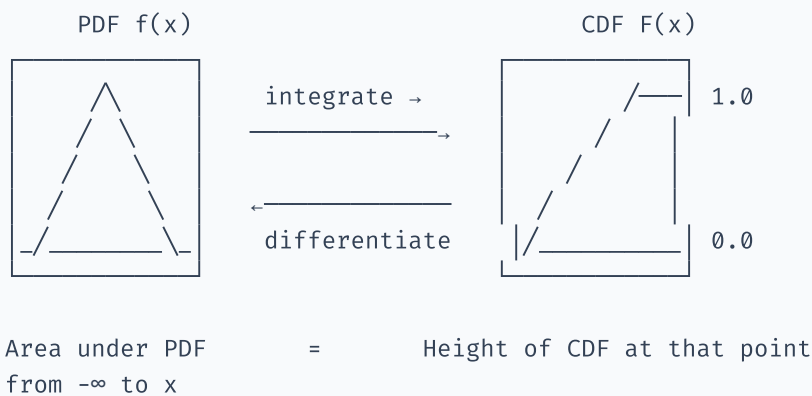
The PDF and CDF are two sides of the same coin — they are related through calculus operations.

8.1 The Fundamental Relationship

PDF ↔ CDF CONVERSION

PDF → CDF: $F(x) = \int_{-\infty}^x f(t) dt$ (Integrate the PDF) CDF → PDF: $f(x) = dF(x)/dx$ (Differentiate the CDF)

Visual Relationship



8.2 Analogous Relationship for Discrete Variables

OPERATION	DISCRETE	CONTINUOUS
PMF/PDF → CDF	$F(x) = \sum p(x_i) \text{ for } x_i \leq x$ (Summation)	$F(x) = \int_{-\infty}^x f(t) dt$ (Integration)
CDF → PMF/PDF	$P(X = x) = F(x) - F(x^-)$ (Difference)	$f(x) = dF(x)/dx$ (Differentiation)

Key Takeaway for ML

In practice, you rarely compute integrals by hand. Libraries like `scipy.stats` provide both the PDF (`.pdf()`) and CDF (`.cdf()`) for any named distribution. Understanding their relationship helps you reason about data distributions, compute percentiles, and set thresholds for classification.

9 • Density Estimation — Parametric vs Non-Parametric

Density estimation is the problem of reconstructing the underlying probability distribution from a finite set of observed data points. This is one of the most fundamental tasks in statistics and ML.

9.1 The Problem

Given observed data $\{x_1, x_2, \dots, x_n\}$, what is the true underlying PDF $f(x)$ that generated this data?

9.2 Two Approaches

Parametric Density Estimation

Assumption: Data follows a specific known distribution (e.g., Normal, Exponential, Poisson).

Task: Estimate only the *parameters* (μ , σ for Normal).

Pros:

- Efficient — needs fewer data points
- Smooth, analytically tractable
- Easy to interpret

Cons:

- If assumption is wrong, estimate is biased
- Inflexible — can't capture multimodal or unusual shapes

Example: Fitting $\hat{\mu} = \bar{X}$ and $\hat{\sigma} = s$ to assume data is Normal

Non-Parametric Density Estimation

Assumption: None! Let the data "speak for itself."

Task: Estimate the entire PDF shape from data.

Pros:

- Flexible — captures any distribution shape
- Works for multimodal, skewed, or arbitrary distributions
- No wrong assumption possible

Cons:

- Needs more data to be accurate
- Sensitive to hyperparameters (bandwidth)
- Computationally heavier

Examples: Histogram, KDE

When to Use Which?

- **Parametric:** When you have domain knowledge that data follows a specific distribution (e.g., measurement errors are typically Normal)
- **Non-parametric:** During exploratory data analysis (EDA), when you don't know the distribution shape, or when data is multimodal/complex

10 • Kernel Density Estimation (KDE)

KDE is the most popular non-parametric density estimation method. It produces a smooth, continuous estimate of the PDF by placing a kernel (small bump) on each data point and summing them up.

10.1 The Intuition

1. **Place a kernel** (a small bell curve or other symmetric shape) centered on *each* data point
2. **Sum all kernels** to get a combined curve
3. **Normalise** so the total area under the curve equals 1

10.2 The Formula

KDE FORMULA

$\hat{f}(x) = (1 / n \cdot h) \cdot \sum_i K((x - x_i) / h)$ where: n = number of data points h = bandwidth (smoothing parameter) $K(\cdot)$ = kernel function (usually Gaussian) x_i = individual data points

10.3 Common Kernel Functions

KERNEL	SHAPE	FORMULA $K(u)$	USAGE
Gaussian	Bell curve	$(1/\sqrt{2\pi}) \cdot e^{-(u^2/2)}$	Most common, default in most libraries
Epanechnikov	Parabolic	$\frac{3}{4}(1 - u^2)$ for $ u \leq 1$	Theoretically optimal (MSE-minimising)
Uniform (Box)	Rectangle	$\frac{1}{2}$ for $ u \leq 1$	Simplest, but not smooth
Triangular	Triangle	$(1 - u)$ for $ u \leq 1$	Simple and smooth

10.4 The Bandwidth Parameter (h) — Most Critical

Bandwidth controls the **width** of each kernel, and thus the **smoothness** of the estimated PDF.

 Small Bandwidth ($h \rightarrow 0$)

- Very narrow kernels $\rightarrow$ spiky, noisy curve
- **Overfitting** — captures noise, not signal
- High variance, low bias
- Shows every minor fluctuation

 Large Bandwidth ($h \rightarrow \infty$)

- Very wide kernels $\rightarrow$ overly smooth curve
- **Underfitting** — hides real structure
- Low variance, high bias
- May miss multiple modes/peaks

 The Bias-Variance Trade-off in KDE

The bandwidth selection is a classic **bias-variance trade-off**:

- Too small $h \rightarrow$ High variance (overfitting)
- Too large $h \rightarrow$ High bias (underfitting)
- Optimal $h \rightarrow$ Balances both

Silverman's Rule of Thumb: $h = 1.06 \cdot \sigma \cdot n^{(-1/5)}$ — a good default for roughly normal data.

11 • Histogram vs KDE

Both histograms and KDE are used for non-parametric density estimation, but they differ fundamentally in their approach and output quality.

FEATURE	HISTOGRAM	KDE
Nature	Discrete (binned rectangles)	Continuous (smooth curve)
Output	Step function — discontinuous	Smooth function — differentiable
Key Parameter	Number of bins / bin width	Bandwidth (h) + kernel choice
Sensitivity	Sensitive to bin edges & starting point	Sensitive to bandwidth
Flexibility	Limited — shape depends heavily on bins	High — smooth, can use different kernels
Edge Effects	Sharp cutoffs at boundaries	Bleeds beyond data range (can be fixed)
Computation	Very fast ($O(n)$)	Slower ($O(n^2)$ naively, but fast with FFT)
Best For	Quick look at distribution shape	Publication-quality density estimates

💡 The Histogram is a Special Case of KDE

A histogram can be viewed as KDE with a **Uniform (box) kernel** where bandwidth equals the bin width. KDE generalises this idea by using smoother kernels and centering them on actual data points rather than on bin edges.

11.1 When to Use Which

- **Histogram:** Quick EDA, initial data exploration, when speed matters
- **KDE:** When you need a smooth density estimate, comparing distributions, presenting results
- **Both together:** Best practice — overlay KDE on a histogram for maximum information

12 · Python Implementation — Complete Code

12.1 PMF — Discrete Random Variable

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

# — PMF of a Fair Die —
x_die = np.arange(1, 7)
pmf_die = np.ones(6) / 6      # uniform PMF: 1/6 each

plt.figure(figsize=(8, 4))
plt.stem(x_die, pmf_die, linefmt='#3b82f6', markerfmt='o', basefmt='k')
plt.title('PMF — Fair Die Roll', fontsize=14, fontweight='bold')
plt.xlabel('Outcome (x)')
plt.ylabel('P(X = x)')
plt.ylim(0, 0.3)
plt.tight_layout()
plt.show()

# — PMF using scipy — Binomial(n=10, p=0.5) —
n, p = 10, 0.5
x_binom = np.arange(0, n+1)
pmf_binom = stats.binom.pmf(x_binom, n, p)

plt.figure(figsize=(8, 4))
plt.bar(x_binom, pmf_binom, color='#8b5cf6', edgecolor='white')
plt.title('PMF — Binomial(n=10, p=0.5)', fontsize=14, fontweight='bold')
plt.xlabel('Number of Successes')
plt.ylabel('P(X = x)')
plt.tight_layout()
plt.show()
```

12.2 PDF — Continuous Random Variable

```
# — PDF of Normal Distribution —
x = np.linspace(-4, 4, 1000)
pdf_norm = stats.norm.pdf(x, loc=0, scale=1)    #  $\mu=0, \sigma=1$ 

plt.figure(figsize=(8, 4))
plt.plot(x, pdf_norm, color='#3b82f6', linewidth=2)
plt.fill_between(x, pdf_norm, alpha=0.2, color='#3b82f6')
plt.title('PDF — Standard Normal Distribution N(0, 1)', fontsize=14)
plt.xlabel('x')
plt.ylabel('f(x) — Density')
plt.tight_layout()
plt.show()

# — Probability over an interval —
prob = stats.norm.cdf(1) - stats.norm.cdf(-1)
print(f"P(-1 ≤ x ≤ 1) = {prob:.4f}")    # ≈ 0.6827 (68.27%)
```

12.3 CDF — Both Discrete & Continuous

```
# — Discrete CDF — Binomial —
cdf_binom = stats.binom.cdf(x_binom, n, p)

# — Continuous CDF — Normal —
cdf_norm = stats.norm.cdf(x, loc=0, scale=1)

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Discrete CDF (step function)
axes[0].step(x_binom, cdf_binom, where='mid', color='#8b5cf6', linewidth=2)
axes[0].set_title('CDF — Binomial(10, 0.5)', fontsize=13)
axes[0].set_xlabel('x'); axes[0].set_ylabel('F(x) = P(X ≤ x)')

# Continuous CDF (smooth S-curve)
axes[1].plot(x, cdf_norm, color='#3b82f6', linewidth=2)
axes[1].set_title('CDF — Normal(0, 1)', fontsize=13)
axes[1].set_xlabel('x'); axes[1].set_ylabel('F(x) = P(X ≤ x)')

plt.tight_layout()
plt.show()
```

12.4 KDE — Kernel Density Estimation

```
import seaborn as sns

# Generate sample data (mixture of two normals)
np.random.seed(42)
data = np.concatenate([
    np.random.normal(loc=2, scale=0.8, size=300),
    np.random.normal(loc=6, scale=1.2, size=200)
])

fig, axes = plt.subplots(1, 3, figsize=(16, 4))

# 1. Histogram only
axes[0].hist(data, bins=30, density=True, color='#94a3b8', edgecolor='white')
axes[0].set_title('Histogram Only')

# 2. KDE only
sns.kdeplot(data, ax=axes[1], color='#3b82f6', linewidth=2, fill=True, alpha=0.3)
axes[1].set_title('KDE Only')

# 3. Histogram + KDE overlay (best practice!)
axes[2].hist(data, bins=30, density=True, color='#94a3b8', edgecolor='white', alpha=0.7)
sns.kdeplot(data, ax=axes[2], color='#ef4444', linewidth=2)
axes[2].set_title('Histogram + KDE')

plt.suptitle('Comparing Density Estimation Methods', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()
```

12.5 Bandwidth Effect on KDE

```
# Demonstrate bandwidth effect
fig, axes = plt.subplots(1, 3, figsize=(16, 4))
bandwidths = [0.1, 0.5, 2.0]
titles = ['h=0.1 (Undersmoothed)', 'h=0.5 (Good)', 'h=2.0 (Oversmoothed)']

for ax, bw, title in zip(axes, bandwidths, titles):
    ax.hist(data, bins=50, density=True, color='#e2e8f0', edgecolor='white')
    sns.kdeplot(data, bw_adjust=bw, ax=ax, color='#3b82f6', linewidth=2)
    ax.set_title(title, fontsize=11, fontweight='bold')

plt.suptitle('Effect of Bandwidth on KDE', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()
```

12.6 Using `scipy.stats` for Any Distribution

```
# — Complete workflow with scipy.stats —
dist = stats.norm(loc=100, scale=15)      # IQ scores:  $\mu=100$ ,  $\sigma=15$ 

# PDF value at a point
density_at_100 = dist.pdf(100)           #  $f(100)$  — peak density

# CDF — cumulative probability
prob_below_130 = dist.cdf(130)           #  $P(X \leq 130) \approx 0.9772$ 

# Survival function (complement of CDF)
prob_above_130 = dist.sf(130)            #  $P(X > 130) \approx 0.0228$ 

# Percent point function (inverse CDF = quantile)
iq_95th = dist.ppf(0.95)                 # 95th percentile  $\approx 124.67$ 

# Interval probability
prob_85_to_115 = dist.cdf(115) - dist.cdf(85)  #  $P(85 \leq X \leq 115) \approx 0.6827$ 

print(f"P(IQ  $\leq$  130) = {prob_below_130:.4f}")
print(f"P(IQ  $>$  130) = {prob_above_130:.4f}")
print(f"95th percentile IQ = {iq_95th:.2f}")
print(f"P(85  $\leq$  IQ  $\leq$  115) = {prob_85_to_115:.4f}")
```

13 · Quick Reference Card

Session 40 — PDF, PMF & CDF Cheat Sheet

Random Variables

- **Discrete:** Countable values → use PMF
- **Continuous:** Uncountable values in a range → use PDF
- **CDF** works for both: $F(x) = P(X \leq x)$

The Three Functions

- **PMF:** $P(X = x) = p(x)$, $\sum p(x) = 1$ (discrete only)
- **PDF:** $f(x)$ = density (NOT probability!), $\int f(x)dx = 1$, $f(x)$ CAN be > 1
- **CDF:** $F(x) = P(X \leq x)$, range $[0, 1]$, non-decreasing

Key Conversions

- PDF → CDF: **Integrate** → $F(x) = \int f(t)dt$
- CDF → PDF: **Differentiate** → $f(x) = dF/dx$
- $P(a \leq X \leq b) = F(b) - F(a) = \int_a^b f(x)dx$

Density Estimation

- **Parametric:** Assume a distribution, estimate parameters (μ, σ)
- **Non-parametric:** Let data speak — Histogram, KDE
- **KDE:** $\hat{f}(x) = (1/nh) \sum K((x-x_i)/h)$ — bandwidth h is critical
- Small h → overfitting, Large h → underfitting

Python Quick Reference

- `stats.norm.pdf(x,  $\mu$ ,  $\sigma$ )` — PDF value at x
- `stats.norm.cdf(x,  $\mu$ ,  $\sigma$ )` — $P(X \leq x)$
- `stats.norm.ppf(q,  $\mu$ ,  $\sigma$ )` — Inverse CDF (quantile)
- `stats.norm.sf(x,  $\mu$ ,  $\sigma$ )` — $P(X > x) = 1 - \text{CDF}$
- `sns.kdeplot(data, bw_adjust=h)` — KDE plot
- `sns.histplot(data, kde=True)` — Histogram + KDE overlay

 What's Next? → Normal Distribution (Session 41)

The next session dives deep into the **Normal (Gaussian) Distribution** — the most important distribution in statistics and ML. Topics include the bell curve, the 68–95–99.7 rule, standard normal distribution, Z-scores, and the Central Limit Theorem.